Digital Cardiac Signal Processing and Analysis (ECG and PPG)

Technical Report of Personal Project: Heuristic Evaluation & Multimodal Fusion (v1) vs. Recalibrated Pan-Tompkins Algorithm Implementation (v2)

Domain: Digital Signal Processing (DSP) & Bioengineering | Sampling Frequency: 1000 Hz

1. EXECUTIVE SUMMARY

This personal engineering project investigates and develops a robust software framework to process biopotential Electrocardiography (ECG) and optical Photoplethysmography (PPG/PHOTO) signals for extracting instantaneous heart rate measured in beats per minute (BPM). The work was executed in two consecutive phases:

- **Version 1 (Exploratory Phase):** Empirical signal evaluation under clean and noisy conditions, critical assessment of signal interpolation routines (`interp1` and `downsample`), and testing of a weighted multimodal fusion heuristic.
- **Version 2 (Rigorous Standard Phase):** Complete algorithm redesign through the adaptation and implementation of the classic **Pan-Tompkins** QRS detector, recalibrating its digital filter transfer functions specifically for signals sampled at **1000 Hz (Fs = 1000 Hz)**.

2. VERSION 1: BIOSIGNAL ANALYSIS, INTERPOLATION & MULTIMODAL FUSION

2.1. Individual Signal Behavior Under Noise Artifacts

Multiple time intervals were inspected to analyze the stability of instantaneous heart rate estimations when subjected to muscular, electromyographic, or movement noise:

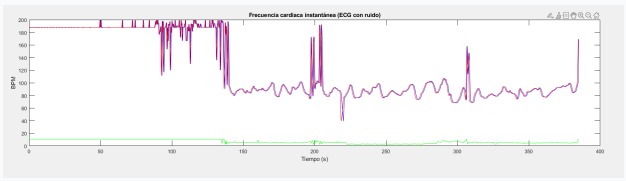


Figure 1. Clean ECG segment (140s - 150s), displaying a stable heart rate between 80 and 90 BPM.

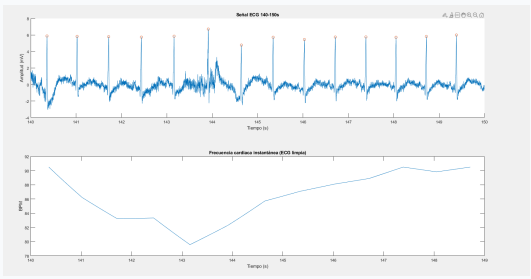


Figure 2. Noisy ECG segment (290s - 320s), showing spurious artifact peaks spiking above 140 BPM.

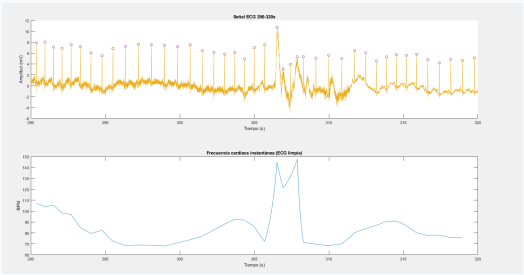


Figure 3. Clean FOTO/PPG segment (150s - 180s), smoothly varying between 80 and 100 BPM.

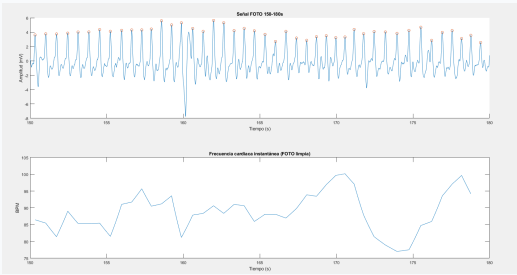


Figure 4. Saturated FOTO/PPG segment (190s - 210s) caused by ADC clipping and excess ambient light.

2.2. Critical Assessment of Data Interpolation (interp1 / downsample)

Interpolation was initially proposed to smooth the graphical representation and align time vector dimensions. However, code evaluation revealed that it had no physical effect:

EVALUATED INTERPOLATION CODE SNIPPET

```
new_t_2 = downsample(t_ecg, 1);
HRV_f_2 = interp1(t_ecg(locs_ecg(1:end-1)), HRV, new_t_2);
```

```
new_t_2 = downsample(t_ecg,1);
HRV_f_2 = interp1(t_ecg(locs_ecg(1:end-1)), HVR, new_t_2);
```

Figure 5. Visual verification: interpolated signal (red) and raw signal (blue) are perfectly identical.

Because `downsample(t_ecg, 1)` uses a downsampling factor of 1, the original sampling density of 1000 samples/s remains unchanged. Halving the rate (e.g., down to 500 Hz) would result in a loss of valuable temporal information. Consequently, interpolation was discarded as redundant.

2.3. Multimodal Weighted Fusion (ECG + PPG) & Structural Flaws

A weighted average heuristic was tested by assigning a weight factor alpha = 0.7 to prioritize the sensor channel with the higher estimated Signal-to-Noise Ratio (SNR):

WEIGHTED FUSION FORMULA

$$HR_{fusion} = \alpha \cdot BPM_{ECG} + (1 - \alpha) \cdot BPM_{PPG} \quad (\text{where } \alpha = 0.7)$$

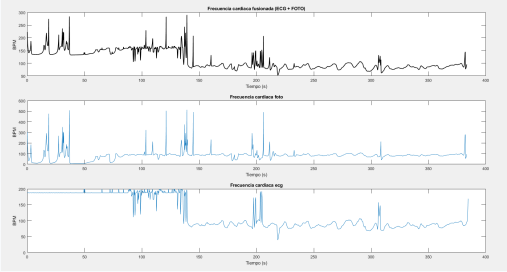


Figure 6. Comparative instantaneous heart rate traces: PPG, ECG, and the fused signal output.

Critical Limitations of Heuristic Fusion in v1

Although the global mean of the fused signal yielded an apparent value of **97.06 BPM**, the method suffers from two fundamental engineering drawbacks:

1. **Physiological Phase Lag (Pulse Transit Time - PTT):** R-peaks in ECG lead the peripheral arterial pulse peaks in PPG by a variable time delay.
2. **Cross-Contamination of Noise:** Artifacts in one channel distort the weighted average even when the secondary sensor signal is entirely clean.

3. VERSION 2: PAN-TOMPKINS ALGORITHM RECALIBRATED TO 1000 HZ

To address the vulnerabilities of Version 1, the formal architecture of the **Pan-Tompkins** QRS detector [1] was implemented. All digital filter stages were recalibrated for a sampling frequency of **1000 Hz (Fs = 1000 Hz)** instead of the standard 200 Hz.



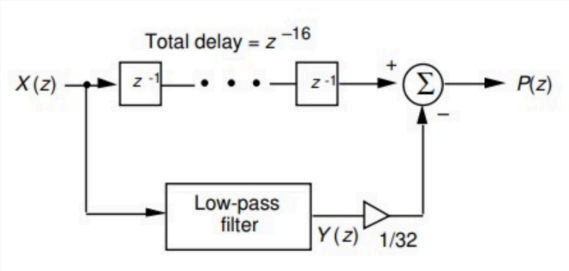


Figure 7. Functional block diagram of the high-pass filter with z^{-16} delay line for DC baseline wander suppression.

3.1. Adapted Digital Transfer Functions (Scaling Factor K = Fs / 200 = 5)

Because the sample rate is 5 times higher than original benchmark specifications, the z-domain poles and zeros were time-scaled accordingly:

ADAPTED LOW-PASS FILTER (CUTOFF ~15 HZ)

$$H_{lp}(z) = (1 - 2z^{-30} + z^{-60}) / (1 - 2z^{-1} + z^{-2})$$

ADAPTED HIGH-PASS FILTER (DC SUPPRESSION)

$$H_{hp}(z) = (1 - z^{-160}) / (1 - z^{-1})$$

ADAPTED DERIVATIVE FILTER

$$H_{der}(z) = (1 / 40) \cdot (2 + z^{-5} - z^{-15} - 2z^{-20})$$

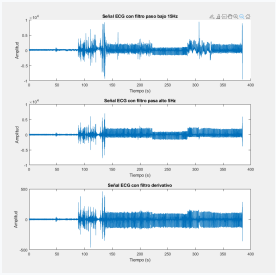


Figure 8. Output waveforms from low-pass (15 Hz), high-pass (5 Hz), and derivative filter stages.

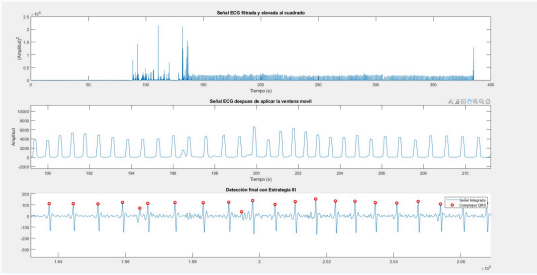


Figure 9. Squared signal, moving window integration (MWI), and final R-peak detection markers.

3.2. Moving Window Integration (MWI) & Adaptive Thresholding

The moving window integration attenuates prominent T-waves to eliminate false beats. Following integration, dual adaptive thresholds dynamically update signal and noise levels to maintain high detection accuracy despite amplitude variations.

4. COMPARATIVE RESULTS & DISCUSSION

A summary of the mean heart rate metrics computed across all experimental configurations is tabulated below:

Method / Evaluated Channel	Mean Heart Rate (BPM)	Performance Assessment
Clean ECG (Version 1)	**84.00 BPM**	High stability during noise-free intervals.
Noisy ECG (Version 1)	**123.43 BPM**	Severe overestimation caused by false positives.
Clean PPG / FOTO (Version 1)	**54.00 BPM**	Conservative estimation from peripheral pulse.
Noisy PPG / FOTO (Version 1)	**80.57 BPM**	Distorted by sensor saturation artifacts.

Method / Evaluated Channel	Mean Heart Rate (BPM)	Performance Assessment
Weighted Fusion ECG+PPG (v1)	**97.06 BPM**	Conceptually inaccurate due to R-PPG phase lag.
Recalibrated Pan-Tompkins (v2)	**115.90 BPM**	**Superior noise rejection and QRS detection accuracy.**

```
>> codigo_limpio_v2
El valor de la tasa cardiaca media en la señal ECG sin ruido es de 84.00
El valor de la tasa cardiaca media en la señal ECG con ruido es de 123.43
El valor de la tasa cardiaca media en la señal FOTO con ruido es de 80.57
El valor de la tasa cardiaca media en la señal FOTO sin ruido es de 54.00
El valor de la tasa cardiaca media en la señal conjunta con ruido es de 97.06
>> PanTompkins
El ritmo cardíaco medio es de 115.90 BPM
```

Figure 10. Console output confirming the mean heart rate calculated by Pan-Tompkins (115.90 BPM).

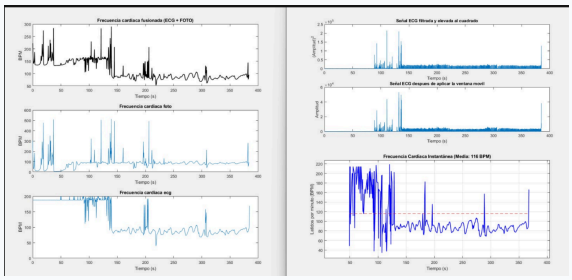


Figure 11. Overall comparison between Version 1 traces and the Pan-Tompkins processed output.

Engineering Conclusion

Recalibrating the Pan-Tompkins algorithm for a 1000 Hz sampling frequency effectively eliminated cutoff frequency misalignment. This bandpass-filtered detection model proved significantly more reliable and robust against power-line and electromyographic interference than the heuristic promediating methods tested in Phase 1.

5. TECHNICAL REFERENCES

1. Jiapu Pan and Willis J. Tompkins, *"A Real-Time QRS Detection Algorithm,"* IEEE Transactions on Biomedical Engineering, vol. BME-32, no. 3, pp. 230-236, March 1985.